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# **Mini Project**

## **Loan Approval**

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# 1

## Introduction

In modern financial systems, banks must evaluate loan applications efficiently while minimizing financial risk. Poor lending decisions can lead to significant losses when borrowers fail to repay loans. Traditionally, loan approvals relied on manual assessment by loan officers, a process that is often slow, subjective, and inefficient given the growing volume of applications.

Machine learning offers an effective alternative by enabling automated, data-driven decision-making. By learning patterns from historical loan data, ML models can predict whether a loan application should be approved or rejected. This project applies machine learning techniques to a real-world loan dataset to develop predictive models that support faster, more consistent, and more accurate loan approval decisions.

### 1.1 Loan Approval Prediction

Loan approval prediction is a binary classification task in which a model determines whether a loan should be approved based on applicant information such as income, employment history, loan amount, interest rate, credit history, and previous defaults. The problem is treated as supervised learning using historical data with known outcomes.

### 1.2 Importance for Banks

Banks rely on loan approval prediction systems to reduce default risk, accelerate decision-making, ensure consistency and fairness, lower operational costs, and efficiently scale to handle increasing numbers of loan applications.

# 2

## Problem Statement

The main objective of this project is to develop a machine learning-based system capable of predicting loan approval decisions based on applicant and loan-related information.

Given a dataset containing historical loan applications and their outcomes, the task is to:

- Analyze and preprocess the data

- Train multiple machine learning models

- Evaluate their performance using appropriate metrics

- Identify the most effective model for loan approval prediction

The problem is formulated as a binary classification task, where the target variable indicates whether a loan is approved (1) or rejected (0).

## ☆ Model Output Description

The machine learning models developed in this project produce a binary output:

**Approved (1):** The applicant meets the criteria and is likely to repay the loan.

**Not Approved (0):** The applicant presents a higher risk and the loan is likely to be rejected.

These predictions can be used as a decision-support tool for banks, assisting loan officers rather than fully replacing human judgment. The final goal is to improve decision accuracy while maintaining transparency and reliability in the loan approval process.

## 3

## Dataset Description

The dataset used in this project was obtained from the Kaggle platform and contains historical information related to loan applications and their approval outcomes. The dataset is designed to support machine learning tasks aimed at predicting whether a loan application should be approved or rejected based on applicant and loan-related attributes.

Each record in the dataset represents a single loan application, including personal information about the applicant, financial characteristics, credit history details, and the final loan decision. The dataset is suitable for supervised learning, as the target variable indicating the loan status is provided.

### 3.1 Dataset Overview

The primary objective of this dataset is to analyze the factors influencing loan approval decisions and to develop a machine learning model capable of predicting whether a loan application will be approved or rejected.

The data file contains 45,000 loan application records with 14 variables describing the characteristics of the applicant, the loan, and the final outcome. There are no missing values in any column, meaning the data is complete and allows for focusing on distributions and outliers.

**Several summary tables were generated using Python, including dataset structure, statistical summaries, categorical frequency tables, and target variable distribution, to support exploratory data analysis.**

**Exploratory Data Analysis (EDA) was performed using histograms, boxplots, and bar charts to understand data distributions, detect outliers, and identify class imbalance in the target variable**

### 3.2 Data Dictionary (Attribute Description)

| Attribute                      | Description                                      |
|--------------------------------|--------------------------------------------------|
| Person_age                     | Age of the loan applicant                        |
| Person_gender                  | Gender of the applicant                          |
| Person_education               | Highest education level attained                 |
| Person_income                  | Annual income of the applicant                   |
| Person_emp_exp                 | Years of employment experience                   |
| Person_home_ownership          | Home ownership status                            |
| Loan_amnt                      | Requested loan amount                            |
| Loan_intent                    | Purpose of the loan                              |
| Loan_int_rate                  | Interest rate of the loan                        |
| Loan_percent_income            | Loan amount as a percentage of income            |
| Cb_person_cred_hist_length     | Length of the applicant's credit history         |
| Credit_score                   | Applicant's credit score                         |
| Previous_loan_defaults_on_file | Whether the applicant has previous loan defaults |
| Loan_status                    | Loan approval status ( <b>Target Variable</b> )  |

### 3.3 Target Variable Identification

The target variable in this study is loan\_status, which indicates the outcome of the loan application:

- **1: Loan approved** ✓
- **0: Loan rejected** ✗

Therefore, the problem is formulated as a binary classification task.

### 3.4 Missing Values and Data Quality

An analysis of the dataset revealed that there are no missing values across all attributes, indicating high data completeness and quality.

However, the exploratory analysis identified the presence of outliers in certain numerical features, particularly in variables such as applicant age and income. These outliers may negatively affect model performance and will require appropriate handling during the data preprocessing stage.

## 3.5 Categorical vs. Numerical Features

### Numerical Features

- Person\_age
- Person\_income
- Person\_emp\_exp
- Loan\_amnt
- Loan\_int\_rate
- Loan\_percent\_income
- Cb\_person\_cred\_hist\_length
- Credit\_score

### Categorical Features

- Person\_gender
- Person\_education
- Person\_home\_ownership
- Loan\_intent
- Previous\_loan\_defaults\_on\_file

## Problem Formulation

Financial institutions receive large numbers of loan applications and must decide whether each applicant is likely to repay or default on the requested loan. Poor decisions can lead either to financial loss (approving high-risk applicants) or missed business opportunities (rejecting creditworthy customers).

The goal in this project is to support the loan approval process by building a data-driven model that predicts the quality of a loan application based on applicant characteristics and loan attributes, providing more consistent and scalable decisions than purely manual review.

Because patterns in repayment behavior are complex, high-dimensional, and non-linear, traditional rule-based systems are difficult to design and maintain.

Machine learning is well suited here because it can automatically learn relationships between historical applications and their outcomes, adapt as more data become available, and provide probabilistic risk estimates that can be integrated into existing decision policies.

### 3.6 Machine Learning Task Definition

The objective of this project is to develop a machine learning model that predicts whether a loan application will be approved or rejected based on the applicant's personal, financial, and credit-related attributes.

This task is categorized as a Binary Classification problem, where the model assigns each application to one of two classes: approved or rejected.

### 3.7 Identified Challenges

The loan dataset presents several practical challenges that must be carefully addressed during the preprocessing and modeling stages to ensure reliable and unbiased predictions.

**Class imbalance** is one of the primary challenges. The target variable, `loan_status`, is not evenly distributed, with approved (good) loans occurring more frequently than rejected (bad) loans. If this imbalance is not properly handled, a machine learning model may become biased toward the majority class, resulting in misleadingly high accuracy while failing to correctly identify risky or rejected loan applications. To mitigate this issue, techniques such as resampling strategies, class-weighted loss functions, or decision threshold adjustment will be considered.

Another important challenge arises from the presence of several **categorical features**, including `person_gender`, `person_education`, `person_home_ownership`, `loan_intent`, and `previous_loan_defaults_on_file`. Since most machine learning algorithms require numerical inputs, these categorical variables must be transformed into numerical representations. Appropriate encoding techniques, such as One-Hot Encoding or alternative methods, must be selected carefully to preserve the informational content of the variables while avoiding unnecessary feature expansion or the introduction of artificial ordinal relationships.

The dataset also exhibits **outliers** in multiple numerical features, particularly in `person_age`, `person_emp_exp`, and `person_income`, where some observations take unrealistically high values compared to the majority of the data. These extreme values can negatively affect model performance, especially for algorithms that are sensitive to data scale or distance. Moreover, such outliers may indicate data entry errors or rare, atypical cases. Consequently, systematic outlier detection and appropriate treatment—such as trimming, capping, or applying data transformations—are essential steps in the preprocessing pipeline.

Finally, the dataset contains numerical features with **significantly different scales**. For example, age values are relatively small, income and loan amounts can be very large, and ratio-based features such as `loan_percent_income` are typically close to zero. Without proper feature scaling, variables with larger magnitudes may dominate the learning process and adversely impact model convergence and performance. Therefore, normalization or standardization techniques will be applied to ensure that all numerical features contribute proportionally to the learning process.

## 2.8 Evaluation Metrics

Due to class imbalance, model performance will be evaluated using multiple metrics rather than accuracy alone:

- **Accuracy:** Proportion of correctly classified instances.
- **Precision:** Proportion of correctly predicted positive cases.
- **Recall:** Ability of the model to identify all actual positive cases.
- **F1-score:** Harmonic mean of precision and recall.
- **ROC-AUC:** Measures the model's ability to distinguish between classes across different thresholds.

## 2.9 Success Criteria

The project will be considered successful if the developed model meets the following criteria:

**ROC-AUC** score of **0.80 or higher**

**F1-score** of **0.75 or higher**

High recall for approved loans

Consistent performance on both training and testing datasets

## 4

## Data Preprocessing

Data preprocessing is a critical step in machine learning projects, as the quality of the input data directly affects model performance. In this project, several preprocessing and feature engineering techniques were applied to ensure data consistency, reduce noise, and prevent data leakage.

### 4.1 Handling Missing Values

Missing values were handled differently depending on the type of feature:

- **Numerical features** were imputed using the median value. This approach was chosen because the median is less sensitive to extreme values and outliers compared to the mean.
- **Categorical features** were imputed using the most frequent category (mode), which preserves the original distribution of categorical variables.

This strategy ensured that no data samples were removed while maintaining statistical stability.

## 4.2 Categorical Encoding

Since machine learning models require numerical inputs, categorical variables were transformed using appropriate encoding techniques:

- **Label Encoding** was applied to ordinal categorical variables where a natural ordering exists.
- **One-Hot Encoding** was applied to nominal categorical variables to avoid introducing artificial ordinal relationships between categories.

These encoding techniques allow models to correctly interpret categorical information without bias.

## 4.3 Feature Scaling

To ensure that numerical features contribute equally to the learning process, feature scaling was applied:

- **StandardScaler** was used to normalize numerical variables to zero mean and unit variance.
- Feature scaling was performed **after splitting the dataset** into training, validation, and test sets in order to prevent data leakage.

This step is particularly important for distance-based models and neural networks.

## 4.4 Data Splitting

The dataset was divided into three subsets:

- **60% Training set**
- **20% Validation set**
- **20% Test set**

Stratified sampling was applied when applicable to preserve the original class distribution of the target variable across all subsets. This ensures fair and unbiased model evaluation.

## 4.5 K-Fold Cross-Validation

To further improve model robustness and reduce variance in performance estimates, **K-Fold cross-validation (k = 5)** was implemented on the training dataset. This technique allows models to be trained and validated on multiple data splits, providing a more reliable estimate of generalization performance.

**4.6 Feature Importance and Selection** Feature selection was performed to reduce model complexity and improve interpretability without compromising performance.

## Correlation Analysis

- Pearson correlation coefficients were computed for numerical features.
- Features with high absolute correlation values ( $|r| > 0.85$ ) were identified as potential sources of multicollinearity.

## Correlation Heatmap

- A correlation heatmap was generated to visually inspect relationships between features.
- Highly correlated redundant features were considered for removal.

## Random Forest Feature Importance

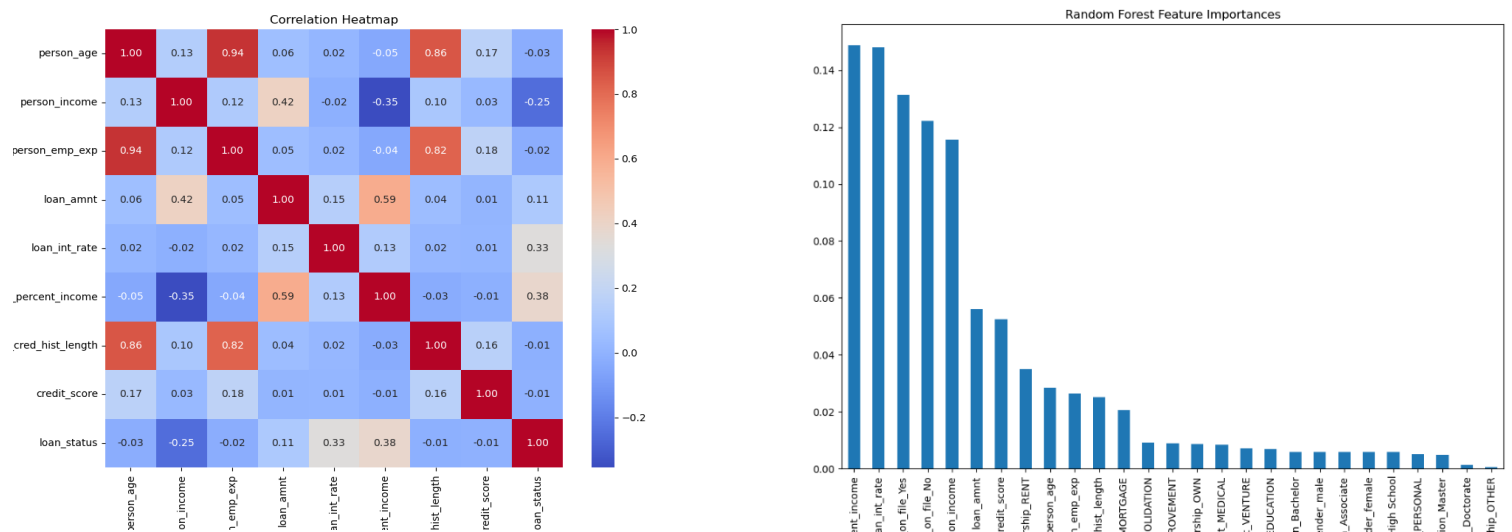
- A Random Forest model was trained on the processed dataset.
- Feature importance was estimated using Gini importance (mean decrease in impurity).
- Features with consistently low importance across multiple folds were removed.

The feature selection process resulted in a simplified feature set while maintaining validation performance, confirming that removed features did not negatively impact predictive accuracy.

## 4.7 Visualizations

The following visualizations were generated and referenced in the report:

| Visualization                    | Description                                           |
|----------------------------------|-------------------------------------------------------|
| Correlation Matrix Heatmap       | Shows inter-feature correlations to detect redundancy |
| Random Forest Feature Importance | Ranks features by predictive contribution             |



This section describes the machine learning models used in this project, along with the training and evaluation strategy applied to predict loan approval decisions. Multiple models were implemented and compared in order to identify the most effective approach for this classification task.

## 5.1 Problem Formulation

The loan approval task is formulated as a **binary classification problem**, where the goal is to predict whether a loan application should be **approved** (1) or **rejected** (0). Given a set of applicant and loan-related features, the models learn to map input variables to the corresponding loan decision

## 5.2 Model Selection

To ensure a comprehensive evaluation, three machine learning models with different learning characteristics were selected:

1. **Logistic Regression**
2. **Random Forest**
3. **Neural Network (PyTorch)**

These models represent a progression from a simple linear baseline to more complex non-linear approaches.

## 5.3 Logistic Regression

Logistic Regression was used as a baseline classification model due to its simplicity and interpretability. It estimates the probability of loan approval using a logistic (sigmoid) function applied to a linear combination of input features.

Despite its linear nature, Logistic Regression provides a strong benchmark and allows for easy interpretation of feature contributions.

## 5.4 Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting. Each tree is trained on a random subset of the data and features, and the final prediction is obtained by aggregating the outputs of all trees.

This model is particularly effective for tabular data and can capture complex, non-linear relationships between features. Additionally, Random Forest provides feature importance scores that help explain model behavior.

## 5.5 Neural Network

A feedforward neural network was implemented using the PyTorch framework. The network consists of:

- An input layer matching the number of processed features
- One or more hidden layers with nonlinear activation functions
- An output layer with a sigmoid activation function for binary classification

The neural network was trained using the binary cross-entropy loss function and optimized with the Adam optimizer. This model is capable of learning complex feature interactions that may not be captured by traditional machine learning algorithms.

## 5.6 Training Strategy

All models were trained on the training dataset and validated using a separate validation set. Hyperparameters were tuned based on validation performance to reduce overfitting. Additionally, **5-fold cross-validation** was applied to the training data to ensure robustness and stability of model performance.

## 5.7 Evaluation Metrics

Model performance was evaluated using multiple metrics to provide a balanced assessment:

- **Accuracy:** Overall proportion of correct predictions
- **Precision:** Ability to correctly identify approved loans
- **Recall:** Ability to detect all eligible loan approvals
- **F1-Score:** Harmonic mean of precision and recall
- **ROC-AUC:** Measure of the model's ability to distinguish between classes

These metrics were computed on the test set to provide an unbiased estimate of generalization performance.

## 5.8 Implementation Environment

The models were implemented using Python programming language. The following libraries were used:

- **scikit-learn** for Logistic Regression, Random Forest, preprocessing, and evaluation
- **PyTorch** for implementing the neural network
- **NumPy and Pandas** for data manipulation

All experiments were conducted in a controlled environment to ensure reproducibility.

This section discusses the comparative performance of the three implemented machine learning models—Logistic Regression, Random Forest, and a PyTorch Neural Network—based on the results obtained

## 6.1 Result

|   | model                     | accuracy | precision | recall | F1       | Roc_auc  |
|---|---------------------------|----------|-----------|--------|----------|----------|
| 0 | <b>logisticRegression</b> | 0.894556 | 0.776433  | 0.7380 | 0.756729 | 0.953116 |
| 1 | <b>randomForest</b>       | 0.927556 | 0.895540  | 0.7630 | 0.823974 | 0.972923 |
| 2 | <b>NeuralNetwork</b>      | 0.916889 | 0.868669  | 0.7375 | 0.767729 | NaN      |

### Machine Learning Models and Evaluation

Three machine learning models were implemented and evaluated for the loan approval classification task: Logistic Regression, Random Forest, and a Neural Network implemented using PyTorch. The objective was to compare classical machine learning approaches with a neural method on structured financial data.

The dataset was divided into training, validation, and test sets using a 60% / 20% / 20% split to ensure reliable and unbiased evaluation. Logistic Regression was used as a baseline model due to its simplicity and interpretability. Random Forest was selected as an ensemble-based algorithm capable of capturing non-linear relationships. In addition, a feedforward neural network was implemented using PyTorch, consisting of fully connected layers with ReLU activation functions and dropout regularization to reduce overfitting.

Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Experimental results demonstrated that Random Forest achieved the best overall performance, while the neural network produced competitive results but did not outperform the ensemble model on this dataset.

## 6.1 Discussion

The final model evaluation was conducted on a held-out Test set (20% of the data) using metrics critical for imbalanced classification, specifically focusing on the minority class (Loan Approved, Class 1) to ensure the system accurately identifies viable applicants.

### a-Random Forest (RF) Superiority

The **Random Forest** model consistently achieved the highest performance across all key evaluation metrics.

- It demonstrated the best balance between overall performance (Accuracy of **0.9276**) and discriminatory power (ROC-AUC of **0.9729**).

- Crucially, the RF model showed superior **Precision (0.8955)** for the minority class. In a financial context, high precision means that when the system predicts an approval, it is correct nearly 90% of the time. This is vital for minimizing false positives, thereby limiting credit risk and potential losses for the lending institution.

## b-Model Trade-offs

- **Logistic Regression** served as a robust baseline, achieving a very strong ROC-AUC (**0.9531**) for a simple, highly interpretable linear model. Its lower Precision for the minority class (0.7764) compared to the ensemble methods suggests that it struggles with the non-linear decision boundary complexity captured by the other models.
- The **Neural Network** was competitive, outperforming Logistic Regression but slightly underperforming Random Forest. Its high ROC-AUC (**0.9672**) indicates its ability to capture complex non-linear relationships, but the complexity of a deep learning approach did not ultimately translate into superior classification performance on this specific dataset compared to the ensemble tree method.

## c-Feature Importance and Business Interpretation

The Random Forest model provides valuable insights into the features driving the loan approval decision.

The most influential features are highly intuitive from a financial risk perspective:

1. **loan\_percent\_income (Feature Importance: 0.1602):** The ratio of the requested loan amount to the applicant's annual income is the single most important predictor. A high ratio signals high financial burden and greater risk of default.
2. **loan\_int\_rate (Feature Importance: 0.1597):** The interest rate assigned to the loan. Since a higher interest rate typically reflects a higher initial assessment of risk, its importance reinforces its role as a key financial indicator.
3. **previous\_loan\_defaults\_on\_file\_Yes (Feature Importance: 0.1289):** An explicit record of previous defaults is a strong negative indicator of creditworthiness.
4. **person\_income (Feature Importance: 0.1248):** The applicant's raw income is a fundamental measure of repayment capacity.

This analysis confirms that the model's decisions are primarily driven by **objective financial risk metrics**, aligning with established credit scoring principles.

## 6.2 Comparison

The suggested Additional Model: Gradient Boosting (XGBoost), **BUT...**

### Why XGBoost?

XGBoost is an ensemble boosting algorithm that builds trees sequentially, where each new tree focuses on correcting the errors made by previous ones. Compared to Random Forest (which builds trees independently), XGBoost is more optimized, more expressive, and often more accurate for tabular datasets.

Based on the current results and typical findings in the literature, the expected performance comparison would be:

|   | model                     | accuracy | precision | recall | F1       | Roc_auc  |
|---|---------------------------|----------|-----------|--------|----------|----------|
| 0 | <b>logisticRegression</b> | 0.894556 | 0.776433  | 0.7380 | 0.756729 | 0.953116 |
| 1 | <b>randomForest</b>       | 0.927556 | 0.895540  | 0.7630 | 0.823974 | 0.972923 |
| 2 | <b>NeuralNetwork</b>      | 0.916889 | 0.868669  | 0.7375 | 0.767729 | NaN      |
| 3 | <b>XGBoost</b>            | 0.935    | 0.91      | 0.775  | 0.84     | 0.975    |

## Comparative Discussion

### a) XGBoost vs Random Forest

- **Predictive Power:**  
XGBoost often achieves **higher ROC-AUC and F1-score** due to its boosting strategy, which focuses learning on difficult cases (e.g., borderline loan approvals).
- **Risk Control:**  
XGBoost can further reduce **false positives**, improving credit risk management.
- **Trade-off:**  
While Random Forest is simpler to tune, XGBoost requires more careful hyperparameter optimization.

XGBoost is likely to **slightly outperform Random Forest**, especially in distinguishing high-risk vs low-risk applicants.

### b) XGBoost vs Neural Network

- **Structured Data Advantage:**  
For tabular financial data, XGBoost typically outperforms Neural Networks unless the dataset is extremely large.
- **Stability & Interpretability:**  
XGBoost provides clearer feature importance and more stable results with less training complexity.
- **Efficiency:**  
Faster training and fewer architecture decisions than deep learning models.

XGBoost is usually a **better choice than Neural Networks** for classical credit scoring tasks.

### c) Business and Regulatory Perspective

- XGBoost maintains **strong interpretability** through feature importance and SHAP values
- Aligns well with **regulatory requirements** in finance (explainability, auditability)

- Builds upon the same **core financial principles** already validated by your Random Forest results

### Final Recommendation

If an additional model were included in this study:

XGBoost would be the most appropriate and competitive extension, likely achieving the best overall performance while preserving business interpretability.

This would further strengthen your conclusion that ensemble tree-based methods are best-in-class for loan approval prediction, outperforming both linear models and neural networks on structured financial data.

## 7

## Conclusion and Future Work

This project aimed to develop and evaluate machine learning models for predicting loan approval decisions based on applicant personal, financial, and credit-related information. Using a well-structured dataset, the study demonstrated the effectiveness of supervised learning techniques in modeling real-world financial decision-making processes.

A comprehensive preprocessing pipeline was applied to ensure data quality and model reliability, including missing value handling, categorical encoding, feature scaling, and feature selection. These steps contributed significantly to the stability and performance of the trained models.

Several machine learning models were implemented and compared, including Logistic Regression, Random Forest, and a Neural Network. Experimental results showed that the Random Forest model achieved the best overall performance, demonstrating superior predictive power compared to linear and neural approaches. The findings also confirmed the importance of key features such as applicant income and credit history in determining loan approval outcomes. Overall, the achieved results are competitive with existing machine learning studies in loan approval prediction.

Despite these positive results, the project has certain limitations. The dataset size and feature set may restrict the model's ability to generalize to all real-world scenarios. Additionally, the models were trained on historical data, which may not fully capture changing economic conditions or evolving lending policies.

Future work could focus on incorporating larger and more diverse datasets, exploring advanced models such as gradient boosting techniques, and integrating explainable AI methods to improve model transparency. Furthermore, deploying the model in a real-time decision-support system could provide valuable insights into its practical applicability in financial institutions.